Introduction

Einstein metrics : Riemannian manifolds (M, g) with

$$Ric(g) = \lambda g \qquad \lambda \in \mathbb{R}.$$

Hard to find "interesting" examples

e.g. non-flat compact (M, g) with $Ric(g) = 0$?

Kähler geometry adds more structure, leads to many examples.

e.g. Zero set $z_0^{n+1} + \dots + z_n^{n+1} = 0$ in $\mathbb{CP}^n$ has Ricci flat metric. (Yau)

Course outline

Part I Yau's theorem.

In Kähler geometry Einstein eq^n can be reduced to scalar PDE for $f : M \longrightarrow \mathbb{R}$:

$$\det\Big(\overbrace{h_{i\bar{j}} + \frac{\partial^2 f}{\partial z^i \partial \bar{z}^j}}^{g_{i\bar{j}}}\Big) = e^{F - \lambda f} \det(h_{i\bar{j}}) \qquad (*)$$

equiv. to $Ric(g) = \lambda g$

where h, F are given, $\lambda \in \mathbb{R}$.

<u>Main PDE thm</u> (Yau) If $\lambda = 0, -1$ then a solution f of $(*)$ satisfies apriori estimate

$$\|f\|_{C^3} \leq C(h, F).$$

$\rightsquigarrow$ Yau's existence thm in cases $Ric(g) = -g$, $Ric(g) = 0$

<u>Key point</u>: Sequence of solutions f_i to $(*)$ cannot blow up if corresponding h_i, F_i are controlled.

Part II

How to take limits of sequences (M_i, g_i) of Riemannian manifolds?

Two extremes:

(A) Fixed manifold M with sequence g_i such that

$$g_i \to g_\infty \quad \text{in } C^k \text{ sense to metric } g_\infty$$

(B) Sequence (M_i, g_i) converges to metric space (Z, d) in Gromov–Hausdorff sense

Two general theorems

Thm 1 (Cheeger, Gromov) If for a $C > 0$

$$|Rm(g_i)| < C, \quad \mathrm{diam}(g_i) < C, \quad \mathrm{vol}(g_i) > C^{-1}$$

then a subsequence converges in sense (A) (with $k=1$)

Thm 2 (Gromov) If $Ric(g_i) > -C g_i$, $\mathrm{diam}(g_i) < C$ then a subsequence converges in sense (B)

Part III Cheeger-Colding theory

<u>Informal statement</u> If $Ric(g_i) > -Cg_i$,

$diam(g_i) < C$, $vol(g_i) > C^{-1}$

then subsequence converges in sense (A) $(k<1)$ away from codimension 2 set.

Part IV Kähler case

<u>Thm</u> (Donaldson-Sun, Liu-Sz.)

Let (M, g_i) be Kähler, polarized (g_i are induced by line bundles)

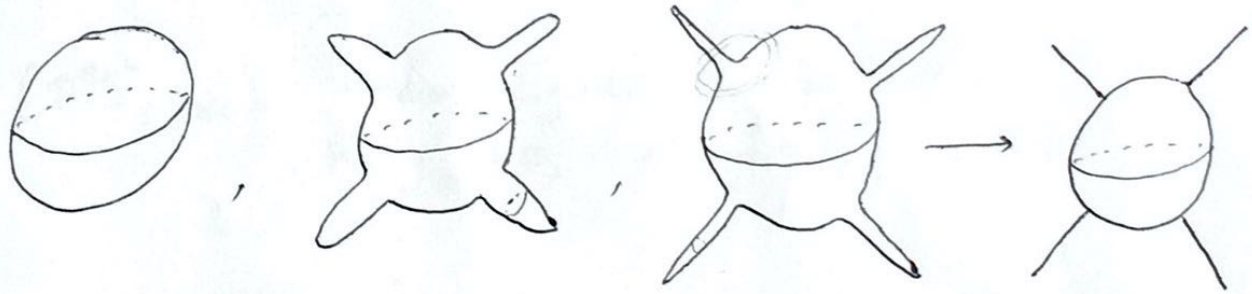
$Ric(g_i) > -Cg_i$, $diam(g_i) < C$, $vol(g_i) > C^{-1}$.

Subsequence converges to (Z, d), homeomorphic to complex projective variety.

Leads to existence of Kähler-Einstein manifolds with $Ric(g) = g$ (Chen-Donaldson-Sun)

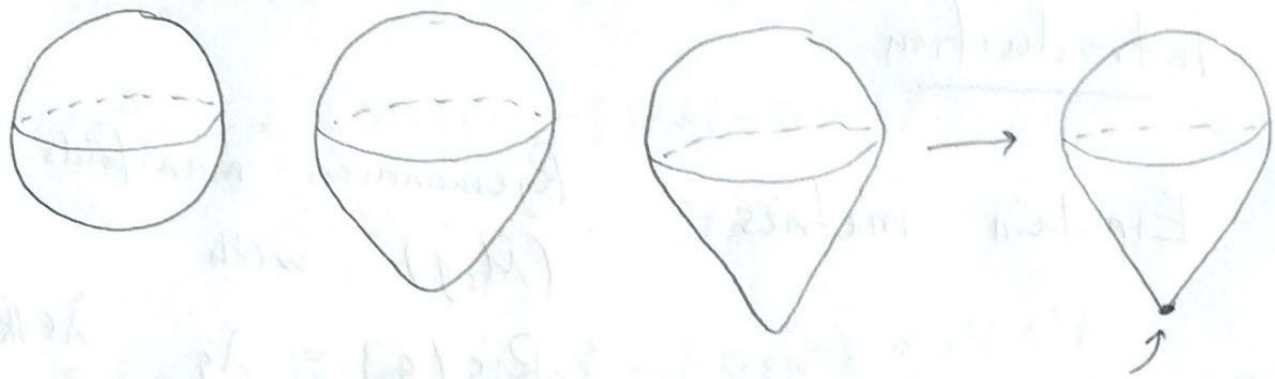
circle

thinner and thinner tori



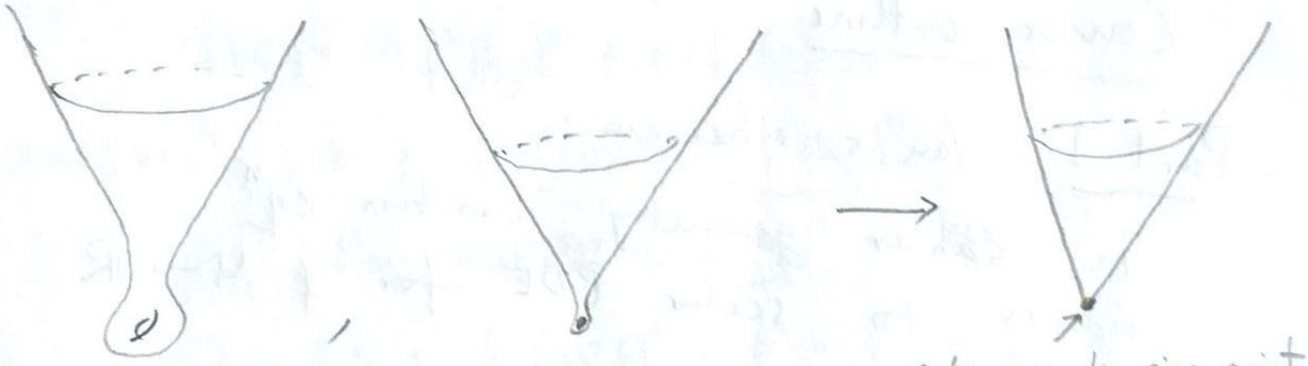
Spheres with bumps

sphere with spikes



Spheres developing cone point Singular point

(T^*S^2, g_{EH}) 4-dim Eguchi-Hanson metric
$Ric(g_{EH}) = 0$, asymptotically conical



g_{EH} $2^{-i} g_{EH}$ Codimension 4 sing. point
flat $\mathbb{C}^2/\mathbb{Z}_2$